Question 8

Q: Given a noisy image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.

Theoretical Concept: Sensor Noise

During image acquisition, noise is introduced by thermal fluctuations (dark current noise) and electronic circuitry (read noise). This is modeled as additive Gaussian noise or Salt-and-Pepper noise.

Minimization: Can be minimized via hardware (cooling the sensor) or software using Spatial Filters like the **Median Filter** (excellent for salt-and-pepper) or **Gaussian Filter**.

💻 OpenCV Python Solution

📋 Copy

```
import cv2  # Import OpenCV for image processing algorithms
import numpy as np                                              # Import Numpy for mat
import matplotlib.pyplot as plt                                 # Import Matplotlib to

# Step 1: Create a clean base image
img = np.full((100, 100), 128, dtype=np.uint8)  # Create a solid gray image array f

# Step 2: Introduce Synthetic Salt-and-Pepper sensor noise
noise = np.random.randint(0, 100, (100, 100))  # Generate random numbers to simulat
img[noise > 95] = 255 # Salt (White)
img[noise < 5] = 0    # Pepper (Black)

# Step 3: Clean the noise using a Median Filter
# The '5' indicates a 5x5 pixel neighborhood used to calculate the mathematical med
clean_img = cv2.medianBlur(img, 5)  # Apply median filter to remove salt-and-pepper

# Step 4: Display the cleaned image
plt.figure(figsize=(8,4))  # Create a new figure window for display
plt.subplot(121), plt.imshow(img, cmap='gray'), plt.title('Sensor Noise')  # Place
```

```
plt.subplot(122), plt.imshow(clean_img, cmap='gray'), plt.title('Median Filter Appl
plt.show()  # Show the final plot window to the user
```

🎯 Expected Output

The output shows a highly speckled, noisy image next to a perfectly restored, flat gray image. The median filter calculates out extreme outliers without blurring actual structure.

🚀 How to Execute & Submit

1. Google Colab Execution:

- Open [Google Colab] and create a New Notebook.
- Click the 📋 **Copy** button on the code block above.
- Paste it into a Colab cell and press **Shift + Enter** to run.

2. Uploading to GitHub:

- In Colab, go to **File** > **Save a copy in GitHub**.
- Authorize your GitHub account.
- Select your Computer Vision Lab repository and click **OK** to commit the code.